

RESEARCH AND ANALYSIS

The environmental impacts of operating an Antarctic research station

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Abstract

We present a life cycle assessment (LCA) of the operation of Casey Station in Antarctica. The LCA included quantifying material and energy flows, modeling of elementary flows, and subsequent environmental impacts. Environmental impacts were dominated by emissions associated with freight operations and electricity cogeneration. A participatory design approach was used to identify options to reduce environmental impacts, which included improving freight efficiency, reducing the temperature setpoint of the living quarters, and installing alternative energy systems. These options were then assessed using LCA, and have the potential to reduce environmental impacts by between 2% and 19.1%, depending on the environmental indicator.

KEYWORDS

Antarctica, Casey Station, industrial ecology, life cycle assessment (LCA), material and energy flow analysis, strategy

1 | INTRODUCTION

1.1 | Background

Antarctica is remote, uninhabited and has some of the earth's most extreme weather conditions. It is critical to the earth's climate systems, with an ice sheet containing a unique record of the earth's climate for over the past 1 million years. Australia was a key architect and one of the 12 original signatories to the 1959 Antarctic Treaty, which has dedicated the Antarctic (which includes the continent of Antarctica) to peace and science (The Secretariat of the Antarctic Treaty, 1959). Between 1959 and 2016, 42 other countries have acceded to the Treaty. There are currently 54 parties in total. Significant research discoveries have occurred on Antarctica, including the ozone layer hole by British researchers in 1985 (Farman, Gardiner, & Shanklin, 1985). It is also a place which has largely been untouched by the impacts of humans, although this is increasingly under pressure due to increased tourism and operational support for science. The Protocol on Environmental Protection (termed the Madrid Protocol; The Secretariat of the Antarctic Treaty, 1991) under the Antarctic Treaty, provides guidelines to signatories for the protection of the Antarctic environment, including those associated with human activities. Despite these guidelines, there is evidence that human activities in the Antarctic are directly impacting the environment (Bargagli, 2008). Indeed, there has been criticism that the requirements for environmental protection in the Madrid Protocol is often qualified by such phrases as "...as far as practical...", which Bennett et al. (2015) suggest are used to "soften environmental obligations."

Research on the impact of human activities has largely focussed on the impacts of local pollutants, including hydrocarbons (e.g., from oil spills), disturbance of flora and fauna, and persistent or bioaccumulative compounds (Bargagli, 2008; Tin et al., 2009). There has been limited research on other environmental burdens associated with human activities in Antarctica, for example, the release of greenhouse gases, or the fate of pollutants arising from energy production and waste incineration. Although the greenhouse gases emitted in Antarctica are small within the global context, the symbolic value of reducing emissions associated with human activities in Antarctica should not be underestimated (Ainley & Tin, 2012). Pertierra, Hughes, Benayas, Justel, and Quesada (2013) investigated the greenhouse gas emissions associated with a remote science station in the South

Shetland Islands in Antarctica, and found that the major drivers for greenhouse gas emissions were associated with transportation of people and cargo.

The Australian Antarctic Division (AAD) operates Australia's three permanent Antarctic stations: Casey, Davis, and Mawson. The AAD has committed to environmental protection under their Environmental Policy (AAD, 2014), including to:

- undertake all reasonable measures including through the setting of objectives and targets to minimize the environmental impacts our activities have, or could have, on the environment;
- reduce pollution, waste, and other human impacts, where practicable, in all our areas of operation;
- use energy efficiently;
- provide environmental education and training for participants in activities supported by the AAD, other visitors to the Antarctic, and the public;
- monitor and report our performance and ensure our environmental decision-making is transparent.

Reducing energy use is an area of ongoing interest for the AAD. Guichard and Steel (1993) conducted a preliminary analysis on energy demand and alternative energy supplies for AAD's Casey and Davis Stations, as well as France's Dumont d'Urville. Similarly, Brown (1998) analyzed the supply and demand for AAD's stations, to investigate the implementation of renewable energy systems. With respect to other anthropological environmental impacts, Wild et al. (2015) found a direct link between the presence of persistent organic pollutants in seawater (including polybrominated diphenyl ethers and perfluoroalkylated substances) and human activities at Casey. O'Brien, Todd, and Kriwoken (2004) measured air emissions associated with Casey's waste incinerator and detected heavy metals (including cadmium, lead, copper, and arsenic). Some of these emissions (copper and arsenic) were considered by the authors as being attributable to the burning of treated timber (a material which has subsequently been banned by the AAD), however no further analysis was conducted on the potential sources for other emissions within the waste stream, nor the fate of these airborne emissions. Yet Antarctic operations play a crucial role in research into the sustainability of the biosphere. In this respect, we argue that more needs to be done to understand how best to improve the environmental sustainability of activities undertaken by nations on the continent. More needs to be done to understand if nations, such as Australia, conduct environmentally conscious Antarctic activities.

1.2 | Study context

Prior to the inception of this study, key personnel from AAD conversed with a researcher from another institution, who had been working with one of our researchers on a life cycle assessment (LCA) project. LCA is a scientific technique used to quantify the potential environmental impacts of a product or service system. The timing of this discussion coincided with the development of Australian Antarctic Science Strategic Plan 2011–12 to 2020–21 (AAD 2011). A challenge identified by the AAD was resolving the apparent conflict between implementing operational support for science activities and associated environmental burdens. For example, AAD identified a need to investigate technologies which would result in zero solid waste discharge at Antarctic stations (AAD 2011). Such technologies exist, but the implementation of these could result in burden shifts to other environmental emissions (e.g., from solid pollutants to air-borne emissions), which may lead to environmentally worse outcomes (Demetrious, Verghese, Stasinopoulos, & Crossin, 2018; Edwards, Othman, Crossin, & Burn, 2017; Procter, Kaplan, & Araujo, 2016). LCA was identified by the AAD, in conversation with our research team, as a potential technique which could improve the understanding of the impacts of AAD's operations in Antarctica. We subsequently commenced an LCA project with the AAD in 2013, linked to Stream 2.3 "Human impacts prevention, mitigation and remediation" of the Australian Antarctic Science Strategic Plan 2011–12 to 2020–21.

To date, there have been no published LCA studies on Antarctic stations. Klee (2005) undertook a material flow analysis on several British Antarctic stations, and extrapolated these to estimate material flows for the entire Antarctic continent. These flows included estimates of key air pollutants, including CO₂. However, these flows were not subject to a life cycle impact assessment, and the process drivers of the air emissions were not reported. In this article, we describe the processes and main outcomes of an LCA on Australia's largest Antarctic research station, Casey Station. The goals of our study were to:

- quantify life cycle environmental impacts of an Antarctic station operation for the first time,
- investigate drivers of impacts, and
- understand the relationship between operations and environmental burdens.

We describe different environmental impact reduction scenarios, including how these were developed and modeled, together with their outcomes. These outcomes are then discussed in the context of informing environmental impact reduction strategies, the development of which we report in an accompanying article (Lockrey, Verghese, Crossin, & Young, 2020). Technical and cost assessments of the described environmental impact reduction scenarios are beyond the scope of this study.

2 | METHODS

2.1 | System under consideration

The AAD selected Casey Station as the site for the LCA. Casey Station is in the Windmill Islands of East Antarctic (located at 66°17' S 110°31' E and herein referred to as Casey). It typically houses 150–160 expeditioners during the summer season (October to February), with approximately 16–20 personnel remaining over the winter period.

The primary energy source for Casey is Special Antarctic Blend (SAB, energy density 43.8 MJ/kg or 35.9 MJ/L lower heating value [LHV]), a diesel fuel formulated for cold conditions. SAB is used for electricity and heat generation, on-site vehicles, and to assist with the incineration of waste.

Electricity at Casey is primarily generated from Caterpillar 3306 direct injection diesel engines, with a nominal capacity of 125 kW, coupled with Stamford alternators and a heat recovery (cogeneration) system. The generators produce 2.77 kWh/L of SAB. Two or three of these generators typically operate at any one time, depending on electricity demand. Two Caterpillar 3412 diesel generators are utilized for backup electricity.

The cogeneration system is integrated with Casey's Site Service Heating Hot Water (SSHHW) system, utilizing a 50% water/50% glycol fluid mixture. The SSHHW system provides hydronic heating, as well as heating for potable water, to key buildings at Casey. The SSHHW utilizes a series of SAB boilers for additional heat generation.

Potable water is produced by a sub-system of the SSHHW system, which pumps hot water into a 50-kg copper melt bell, submerged in a nearby frozen lake. The heat from the hot water delivered through the melt bell melts the surrounding ice. This water is then pumped into tanks, before being treated and distributed to key buildings at Casey. Casey operates its own primary and secondary wastewater treatment (WWT) station, utilizing oil removal, rotating biological contactors, and ultraviolet sterilization systems. This treatment plant handles wastewater from toilets, showers, and sinks (kitchen and bathroom). The treated wastewater is discharged into the ocean via a pipeline, with sewage sludge being returned to Australia (RTA). This system was due to be upgraded to an advanced wastewater system during the 2016/17 season.

Casey is resupplied on an annual basis through freight transfers from an icebreaker (Aurora Australis, although this will be replaced with the RSV Nuyina in late 2020), which utilizes Marine Gas Oil (MGO). Some expeditioners travel to Casey on the Aurora Australis, while others travel on an Airbus A319-115LR, operating out of Hobart, Tasmania. This aircraft lands at Wilkins Aerodrome, located approximately 70 km south-east of Casey. Personnel and freight are typically transferred from Wilkins Aerodrome to Casey via snow vehicles. Fixed-wing aircraft (utilizing Jet A-1 fuel) operate out of Wilkins Aerodrome or the Casey Station Skiway, a small runway closer to Casey. Further logistical support is provided by helicopters (Jet A-1 fuel), which can operate from Wilkins Aerodrome, Casey Station Skiway, or Casey Station.

Solid waste is either incinerated or RTA via the Aurora Australis. Organic waste, including food scraps and cardboard, are incinerated. Composting is not a viable waste management option due to the climate, no end-use, and the potential need to introduce foreign micro-organisms to Antarctica. Additional RTA waste includes batteries, packaging materials, incineration ash, and building materials.

2.2 | Consultation

We used multiple methods to gather data related to the operation of Casey Station and its context. Figure 1 summarizes the key stages and activities of the project.

We engaged a multidisciplinary group of AAD management staff through a series of meetings and co-design workshops throughout the project. Co-design is a way to involve key stakeholders in a series of participatory tasks, to enable their contribution to the final outcome of a project (Technopolis Group, 2015). In this case the project was an LCA and a resultant environmental impact reduction strategy. These staff provided data for modeling, facilitated dialogue with on-station staff, helped identify scenarios for modeling, and provided context with respect to operational decisions, policy, and strategic priorities.

Utilizing internal AAD data, an initial LCA was developed during 2013–2014. This provided an indication of processes and activities that were contributing to identified environmental indicators; highlighted potential issues with data capture and availability; and provided insights into data gaps and the areas of focus for subsequent stages. Two members of the research team conducted a site visit to Casey Station in February 2015, which provided insights into the operation of Casey and additional data. The initial LCA model was subsequently updated, allowing us to further understand material and energy flows on station, and associated operations and expeditioner practices. Co-design workshops, using the Double Diamond methodology (Clune & Lockrey, 2014; Technopolis Group, 2015), were then held with AAD staff to present the outcomes of the LCA, refine data collection, and identify scenarios for modeling. For the Double Diamond methodology, LCA is used to inform the development of the environmental impact reduction strategy. Scenarios were then modeled and presented to the AAD. The remainder of this section describes the LCA modeling.

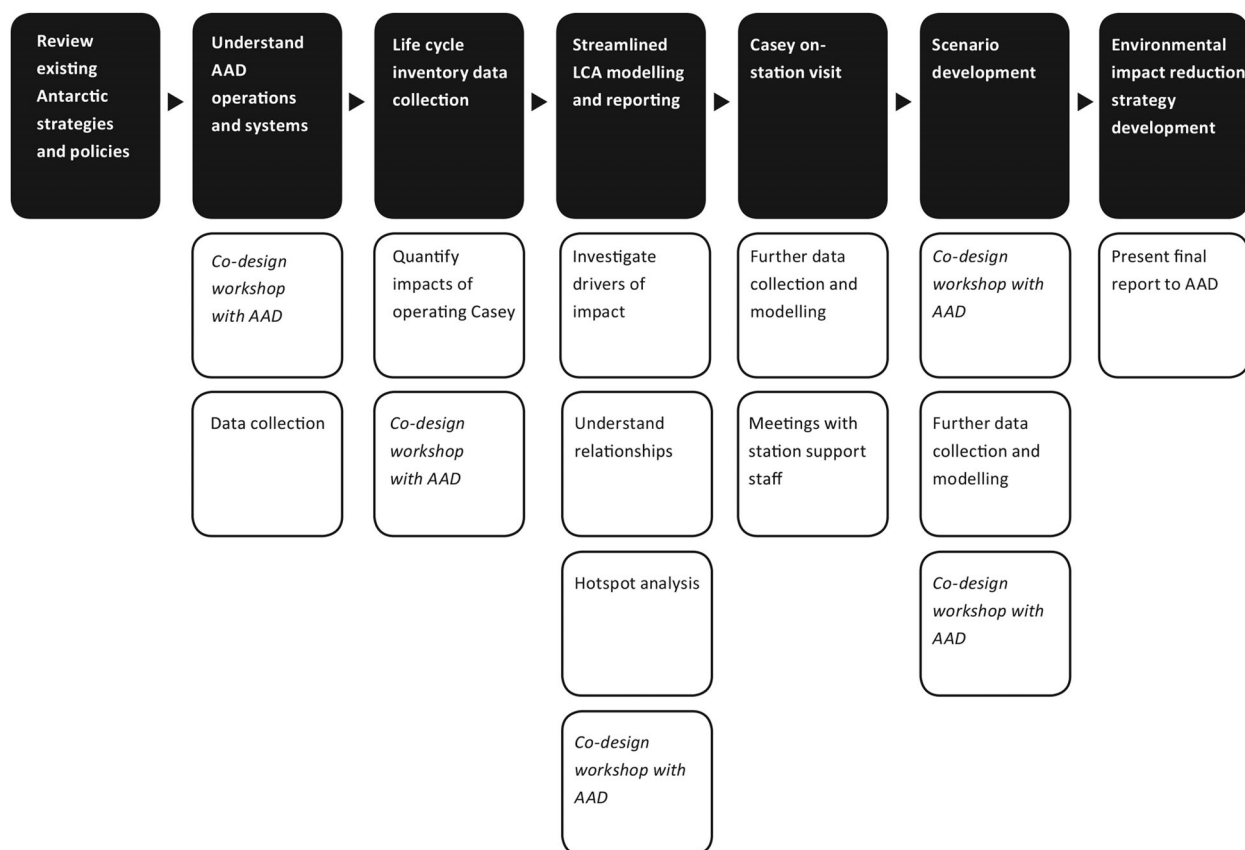


FIGURE 1 Stages of LCA in the last frontier

2.3 | Functional unit

The functional unit (FU) is the unit by which all environmental evaluations are made, and is based on the primary function of the system. We considered several primary functions of Antarctic research stations, including the provision of science, maintaining a presence on the Antarctic continent and operational support for science. We concluded that operational support was most appropriate, which was quantified by the functional unit of the "Operation of Casey Station for one financial year."

2.4 | System boundary

The LCA was limited to processes associated with operating Casey Station, including the supply and combustion of fossil fuels (SAB, Jet A-1, and MGO), the operation of generators and boilers, waste treatment and disposal, and water production and disposal (Figure 2). Other AAD operations, including AAD headquarters in Kingston, Tasmania, were outside the scope of the study. The temporal scope for the LCA was the 2011–2012 financial year, running from July 1, 2011 to June 30, 2012. This year was selected due to data availability, and the steady-state operation of sea voyage and station operations for that period (e.g., there were no unexpected diversions of the Aurora Australis).

2.5 | Life cycle inventory

We collected and collated life cycle inventory (LCI) data relating to material and waste flows for the operation of Casey. Data were collected from the Australian Antarctic Data Centre, or through personal communications. Proxy data from other periods were used when data from this period were unavailable. This data was then coupled with existing LCI databases available at the time of initial modeling, including the Australian Life Cycle Inventory (AusLCI) and ecoinvent 2.2. Inventory tables are provided in Supporting Information S1.

2.6 | Life cycle impact assessment

The LCI flows were then assessed across a range of indicators, Table 1, reflecting indicators which were of interest to the AAD, and which were supported by the LCI. Given the lack of regionally relevant marine aquatic ecological toxicity (ecotoxicity) impact assessment methods, a range of existing methods were utilized.

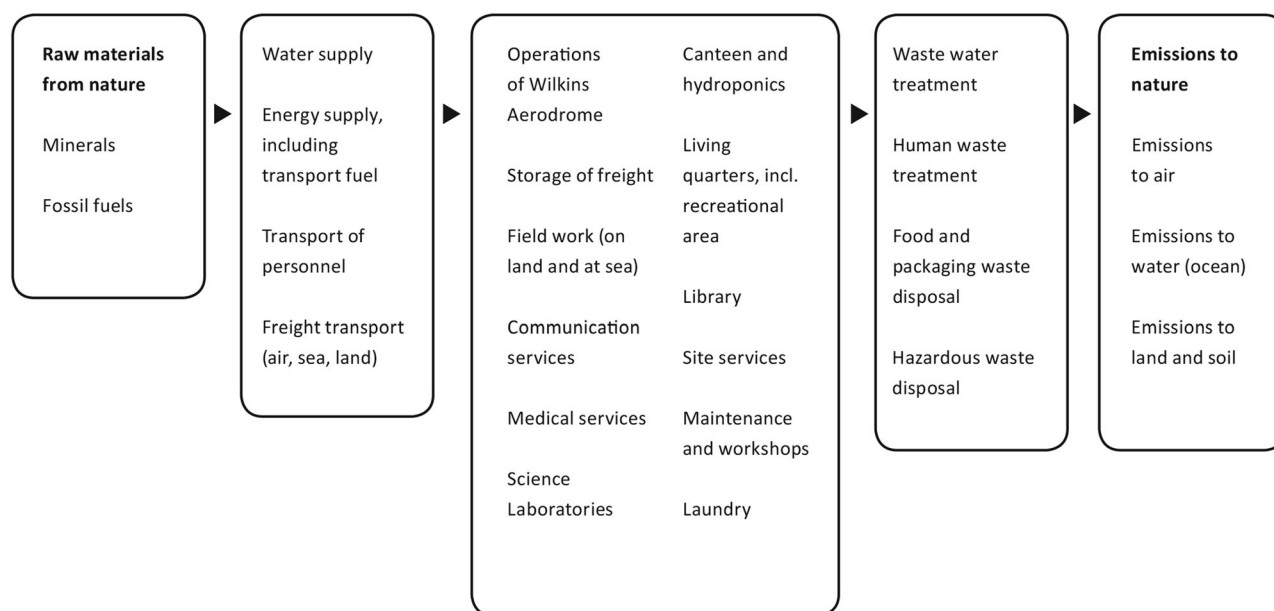


FIGURE 2 System boundary of LCA in the last frontier

TABLE 1 Environmental indicators

Indicator	Unit	Indicator type	Method
Climate change potential	tonne CO ₂ eq	Global, environmental	IPCC Fourth Assessment Report (IPCC, 2007), 100 year time
Cumulative energy demand (CED)	GJ LHV (lower heating value)	Global, inventory	Summation of calorific value of elementary flows
Eutrophication potential (marine)	kg N eq	Local, environmental	ReCiPe (Goedkoop et al., 2009)
Marine ecotoxicity	kg 1,4-DB eq	Local, environmental	ReCiPe (Goedkoop et al., 2009)
Marine ecotoxicity	kg 1,4-DB eq	Local, environmental	CML 2001 (Department of Industrial Ecology [CML-IE], 2016)
Marine ecotoxicity	day	Local, environmental	Lundie, Huijbregts, Rowley, Mohr, and Feitz (2007)

Note: 1,4-DB eq = 1,4 dichlorobenzene equivalents.

3 | BASELINE LCA RESULTS

The baseline life cycle impact assessment results are reported in Table 2.

3.1 | Climate change potential

The climate change potential for the operation and support of Casey Station in 2011/12 was 4,784 tonnes carbon dioxide equivalents (CO₂ eq). The processes contributing to at least 5% of climate change impacts were:

- MGO for Aurora Australis, to transport SAB to Casey, freight and cargo to Casey and waste from Casey (34%)

TABLE 2 Impact assessment results for operation of Casey Station for one financial year: 2011/12 financial year

Impact category	Amount	Unit
Climate change potential	4,784	tonne CO ₂ eq
Cumulative energy demand	59,711	GJ LHV
Marine eutrophication	1,257	kg N eq
Marine ecotoxicity—ReCiPe	17,332	kg 1,4-DB eq
Marine ecotoxicity—CML 2001	3,607	kg 1,4-DB eq
Marine ecotoxicity—AUPLCI	2.48×10^{-7}	day

- SAB for electricity generation (13%) and cogeneration heat (11%)
- SAB for on-site vehicles (13%)
- SAB for boilers (7%)
- Jet A-1 for helicopters and fixed-wing aircraft (12%) and A319-115 LR flights (6%)

Other processes, including waste disposal, contributed to less than 5% of climate change impacts.

3.2 | Cumulative energy demand

The cumulative energy demand (CED) for the operation and support of Casey was 59,711 GJ LHV. Process drivers are similar to that for the climate change potential indicator, with Aurora Australis operations being the largest contributor (37%), followed by electricity generation (14%), on-site vehicles (13%), Antarctic-based flights (12%), cogeneration (11%), boilers (6%), and A319-115 LR flights (6%). All other processes had a total CED of less than 1%.

3.3 | Marine eutrophication

The local marine eutrophication potential for the operation and supply of Casey was 1,257 kg nitrogen equivalents (N eq). The process drivers for this were electricity production (25%), on-site vehicles (23%), Antarctic-based flights (20%), cogeneration (19%), and boilers (11%). These impacts were driven by nitrogen oxide exhaust emissions from these systems. Remaining processes had a cumulative eutrophication impact of less than 1%. These values exclude the potential marine eutrophication impacts of processes occurring outside of Casey Station, such as emissions arising from the use of MGO on the Aurora Australis.

3.4 | Marine ecotoxicity

The process drivers for the local marine ecotoxicity impacts varied, depending on which impact assessment method was used. Regardless of the method, there is a potentially significant contribution from four key processes: electricity generation, cogeneration heat, wastewater disposal, and the use of on-site vehicles. The ecotoxicity impacts for the wastewater treatment plant (WWTP) were driven by heavy metal waste emissions, however the associated elementary flows were based on secondary data, rather than primary data from Casey.

4 | SCENARIO DEVELOPMENT

The following scenarios were identified through the co-design workshops, to explore deployable strategies to reduce environmental impacts at Casey:

1. Reducing the temperature set point in the living quarters (Domestic Red Shed Building)
2. Building benchmarking
3. Utilizing the new icebreaker
4. Installing photovoltaic systems
5. Installing a load-leveling battery

Improvement of the existing building envelopes (e.g., through additional or improved insulation) was not identified by AAD as an immediate environmental reduction option. The methods and results for the identified scenarios are described below.

4.1 | Reducing the temperature setpoint in the living quarters

The temperature of Casey's living quarters (Domestic Red Shed Building) is maintained using a building management system to a setpoint of 19°C. The heat transferred to the outdoor environment depends on several factors, and particularly the external environment. Wind acts as a cooling fluid, removing heat from buildings via convective heat transfer. Given this, it was postulated that there should be a correlation between the heat being delivered through the SSHHW system and a measure of heat transfer (based on the temperature difference between the internal and external environments, the exposed area to the prevailing wind direction, and wind speed). The rate of convective heat transfer, (J/s or W), is governed by the area exposed to a cooling fluid (A_E , m^2), a heat transfer coefficient (h_c , $W/m^2 K$, which varies with the flow rate [speed] of the cooling fluid), and the temperature difference (ΔT , K) between the object and the fluid temperature:

$$\dot{Q} = h_c A_E \Delta T \quad (1)$$

The heat transfer rate can then be integrated over time, t , to estimate the total heat transferred (J):

$$Q = \int_0^t h_c A_E \Delta T dt \quad (2)$$

The exposed area, A_E , was complicated by the multiple external surfaces of the buildings, likely resulting in different air flow regimes (e.g., laminar and/or turbulent), and thus variable values of A_E . As such, the calculated values were based on heat transferred per unit area (Q/A_E), rather than a true value of heat (Q). The heat transfer coefficient (h_c) of air depends on its velocity (v , m/s), and can be approximated by (Moaveni, 2014):

$$h_c = 10.45 - v + 10v^{1/2} \quad (3)$$

The air velocity used in the calculations was based on the daily maximum wind gust, based on data provided by the Australian Bureau of Meteorology. The temperature difference, ΔT , was calculated based on the difference between the setpoint (19°C) and the average daily outside air temperature (°C). The average daily outside temperature range was between -29.1°C and 1.6°C, with a mean of -10.1°C.

The amount of daily heat produced by the cogeneration system was calculated based on the flow rate, heat capacity, and the temperature change of the SSHHW fluid. Logs of the daily flow rates and temperatures for the cogeneration system were provided by the AAD. Both the density (ρ , kg/m³) and the heat capacity (c_p , J/kg °C) of the glycol/water mixture varies with temperature, and were estimated based on regression equations, developed from tables of fluid properties (Engineering Toolbox, 2003):

$$c_p = 3.3947T + 3317.4 \quad (4)$$

$$\rho = 1.0907 - 0.0006T \quad (5)$$

where T (°C) is the average temperature of the glycol mixture entering and leaving the cogeneration system. The daily heat, $Q_{SSH\dot{H}W}$, lost through the SSHHW system was estimated by:

$$Q = c_p \Delta T \dot{V} t \rho \quad (6)$$

where t is the time in 1 day (86,400 s). The amount of daily heat produced by the boiler systems, Q_B , was calculated based on the reported daily volume (provided by the AAD), the LHV of SAB, and an assumed boiler efficiency of 80%. The total heat, Q_T , was the sum of the heat from the SSHHW system ($Q_{SSH\dot{H}W}$) and the boiler system (Q_B).

The outcome of this analysis is reported in Figure 3. As postulated, there was a statistically significant ($p < 0.05$) correlation between the heat transfer per unit area and the daily cogeneration and boiler heat.

The regression relationship above was then used to estimate the heat generation required to achieve a setpoint of 18°C (a reduction of 1°C). Using this, the annual SAB savings associated with this setpoint reduction was estimated to be 6,639 L. Conversely, should the setpoint be increased by 1°C (to 20°C), then it would be expected that SAB use would increase by 6,639 L.

4.2 | Building benchmarking

The electricity per unit area (kWh/m²/month) for key Casey Station buildings were compared with those of a benchmark, based on the United States Energy Information Administration's Commercial Buildings Energy Consumption Survey (United States Energy Information Administration, 2012). Average monthly electricity use for each building at Casey was calculated from data logged by the AAD. The floor areas for the buildings at Casey Station were estimated using scaled building schematics provided by the AAD. To establish a benchmark, the CBECS data were limited to buildings which were free-standing, in cold or very cold climates (no data were available for sub-Arctic climate zones), and had no electrical space heating. Following this, the data were further filtered and categorized based on the primary purpose of the buildings, to (as much as possible) align with the purpose of the different building types at Casey Station. The benchmark matchings were public order and safety buildings for the operations building, non-refrigerated warehouse for Emergency Vehicle Shelter (EVS), Field Store and Workshop, lodgings for the Domestic building, laboratories for the Science building, and refrigerated warehouse for the Green Store. The outcomes of the benchmarking are reported in Figure 4.

Most buildings performed better than the benchmark average, except for the operations building (Ops), EVS, and Workshop. If these three building types were to perform at the average of the equivalent CBECS benchmark, the annual electricity would be reduced by an estimated 178,036 kWh per annum, equivalent to SAB saving of 49,315 L. A reduction in electricity inputs results in an increased demand for SAB use in boilers of 23,538 L per year, resulting in a net reduction in SAB use of 25,777 L per year.

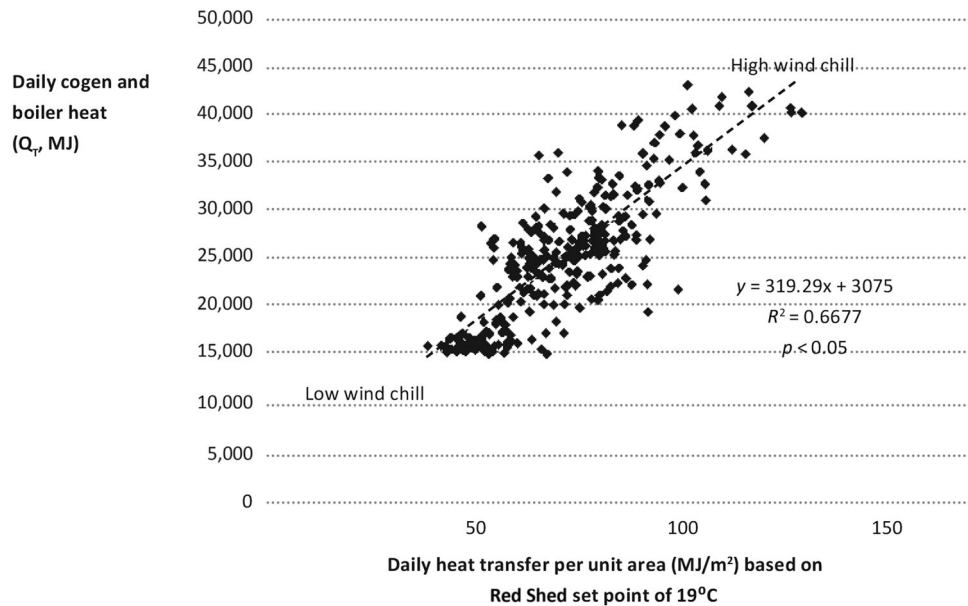


FIGURE 3 Correlation between the daily heat transfer per unit area of the Red Shed with a setpoint of 19°C and the daily heat generated with the cogeneration unit and boilers. Underlying data used to create this figure can be found in Supporting Information S2

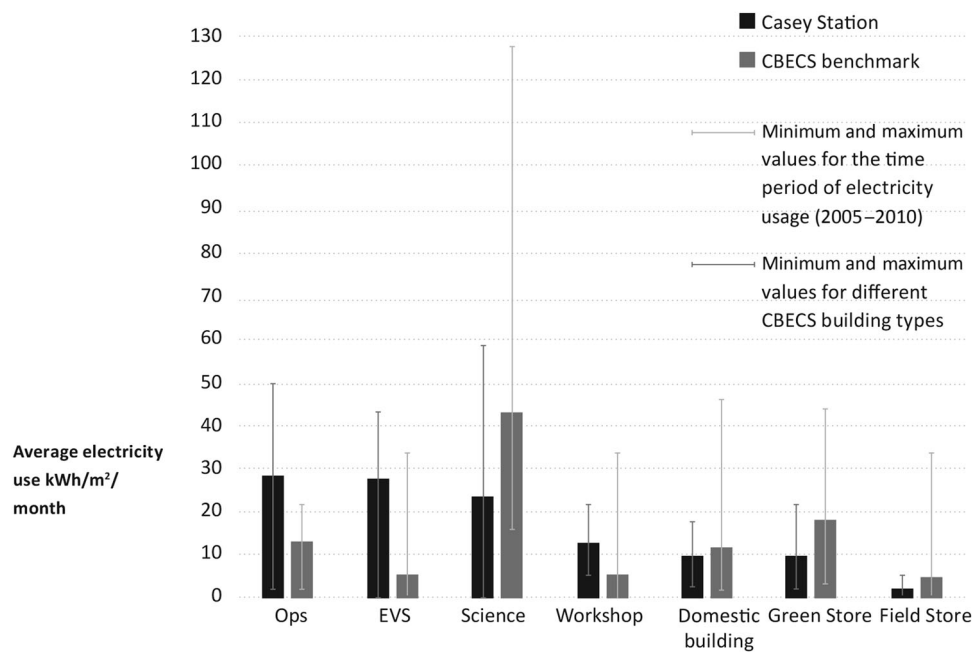


FIGURE 4 Average monthly electricity use per floor area ($\text{kWh}/\text{m}^2/\text{month}$) for different buildings at Casey Station (black), compared with benchmark values from the United States' Commercial Buildings Energy Consumption Survey (CBECS; grey). Axis labels represent the different buildings at Casey (Ops = operations, EVS = Emergency Vehicle Shelter). Error bars for Casey Station represent minimum and maximum values across the 2005–2010 period, while the error bars for the CBECS benchmark represent the range of values for equivalent building types. Underlying data used to create this figure can be found in Supporting Information S2

4.3 | Utilizing the new icebreaker

No voyage-specific fuel estimates were available for the new icebreaker that the AAD is set to deploy, the RSV Nuyina. AAD nominated MGO use rates for the current icebreaker Aurora Australis, and the new icebreaker RSV Nuyina, as being 30 tonne/day and 35 tonne/day, respectively. The MGO use rates (g/km) were estimated based on wayfare data of Aurora Australis voyages between Hobart and Casey for the 2010, 2011, and 2012 calendar years. During this period the Aurora Australis operated for a total of 1,896 hr, resulting in an estimated 2,370 tonne of MGO use. During the same period, the Aurora Australis travelled 23,143 km, equivalent to an MGO use rate of $102.4 \text{ kg}/\text{km}$. The MGO use rate was then

estimated to be 119.5 kg/km. For the purposes of the fuel saving scenario, it was assumed that RSV Nuyina could be used to resupply Casey and another station (Davis) in one voyage. The total distance of this voyage was estimated to be 10,095 km, with resultant MGO fuel use of 1,206 tonne. If this fuel is allocated equally between Casey and Davis (i.e., 603 tonne of MGO fuel use to supply each station), then the MGO fuel savings per year were estimated to be 187 tonne. An alternative resupply scenario would be for the new icebreaker to resupply the station every second year. However, this would mean that the on-station infrastructure warehousing capacity (e.g., for fuel and food) would need to increase twofold, leading to a rebound in other environmental impacts. As such, this alternative resupply scenario was not considered.

4.4 | Installing photovoltaic systems

Hourly global horizontal irradiance (GHI; W/m^2) data for Casey and Wilkins Aerodrome were sourced from Meteonorm (Meteonorm, 2016). The GHI data were processed into kWh of delivered electricity using HOMER Pro (HOMER Energy LLC, 2018), with the following parameters:

- Panel type: 36 cells
- Number of panels: 342 (Casey), 24 (Wilkins Aerodrome)
- Array power (kW): 47.88 (Casey), 3.36 (Wilkins)
- Total panel area: 299.6 m^2 (Casey), 105 m^2 (Wilkins Aerodrome)
- Panel efficiency: 15%
- Performance ratio: 75%
- Albedo (Ground reflection): 70%
- Solar panel tilt: 90° (vertical)
- Solar panel direction: 0° (North)

For Casey, it was assumed that power could be used throughout a year, whilst for Wilkins Aerodrome, it was assumed that a PV system would be utilized during the “summer season,” that is, when science projects take place, from the start of October to the end of the following March. The total electricity savings were 43,148 kWh and 12,065 kWh, for Casey and Wilkins, respectively. This corresponds to 11,951 and 3,342 L of SAB savings for Casey and Wilkins Aerodrome, respectively. For Casey, the reduction in cogenerated heat resulted in an increase in SAB use in the boilers 5,704 L, resulting in a net saving of 6,247 L of SAB per year.

For Wilkins Aerodrome, the SAB is supplied in 9,000 L tanks, which are towed from Casey using a tractor, consuming 1,080 L of SAB for a single return trip. The average SAB use for a year’s operation was calculated to be 63,364 L, delivered using an average of eight tanks. The reduction in SAB requirements would likely result in reducing the tank deliveries from eight to seven, with a subsequent additional SAB saving of 1,080 L. In total, the use of PV at Wilkins Aerodrome is estimated to save 4,422 L of SAB per year.

4.5 | Installing a load-leveling battery

The AAD logged the electrical power (kW) generated by the generator(s) over 1 year (2016) using a 5-s time interval. In order to estimate potential savings associated with battery load leveling, data was processed to represent an average 24-hr period. The AAD data was processed into a representative 24-hr period, using Python script to:

- remove any negative, null, or zero values, or those below a nominal 150 kW threshold (equivalent to 60% loading on 2×125 kW generators),
 - calculate the mean electrical load (kW) for each 5-s time increment for the 365 data points for the year
 - calculate the standard deviation of the load (kW) for each 5-s time increment for the 365 data points for the year, and
 - remove any data points beyond a variation ± 3 standard deviation from the mean for each 5-s time increment for the 365 data points for the year.
- This was performed to remove data which limited further interpretation.

This load data was then processed using Microsoft Excel to calculate incremental kWh generation for each 5-s time increment for the 24-hr period. To determine battery storage charging and discharging throughout this 24-hr period, a discharge efficiency of 90% (Li & Tseng, 2015) and recharge efficiency of 90% (Sbordone, Di Pietra, & Bocci, 2015) were assumed. A threshold power value was calculated to determine the load at which generators are tuned to operate at. If power demand from the station was below this threshold value, energy would be stored. Similarly, if power demand was above the threshold, power would be drawn from the battery to supplement power being delivered from the generators.

The power demand over the representative 24-hr period is reported in Figure 5. The average power over the 24-hr period was calculated to be 190.92 kW. The threshold power was calculated to be 191.64 kW. The saving potential for this system was calculated to be 82.05 kWh, equivalent to 29,946 kWh per year, with subsequent generator set SAB savings of 8,295 L per year. This is offset by an increase in the demand for SAB use in boilers of 3,959 L per year.

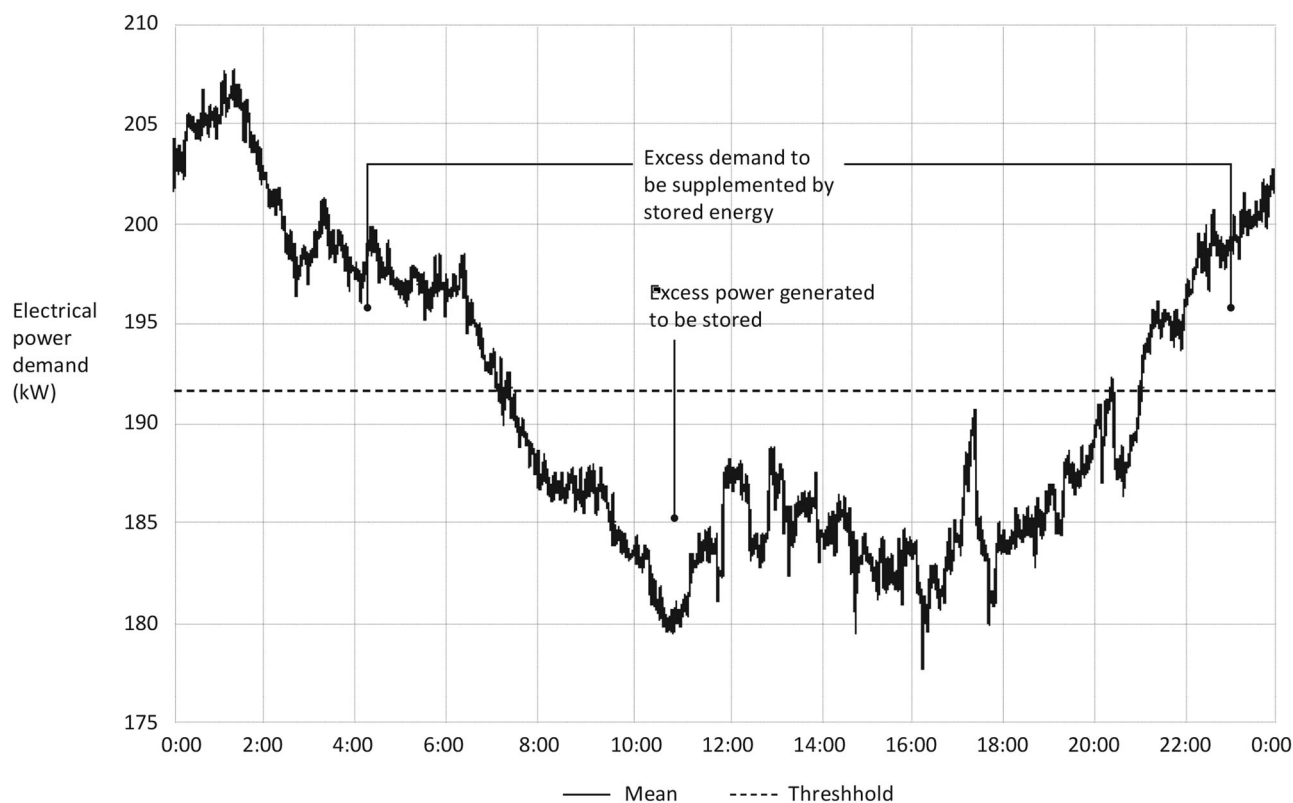


FIGURE 5 Electrical power demand for 24-hr period (average for 2016). The threshold power is 191.6 kW (dashed line) and represents the optimum power generation for the generator sets. If power demand is below this threshold, then excess energy is stored in batteries. If power demand is above this threshold, then power is supplemented from the stored energy. Underlying data used to create this figure can be found in Supporting Information S2

TABLE 3 Impact assessment results for Casey Station scenarios. Results are estimates for the operation of Casey Station for 1 year

Impact category	Unit	Baseline	Temperature setpoint reduction	Building benchmarking	New icebreaker	Solar at Casey Station	Solar at Wilkins Aerodrome	Load leveling battery	Combined scenarios
Climate change potential	tonne CO ₂ eq	4,784	4,766	4,709	4,046	4,766	4,772	4,772	3,909
	Saving		0.4%	1.6%	15.4%	0.4%	0.3%	0.3%	18.6%
Cumulative energy demand	GJ LHV	59,711	61,313	59,489	58,829	50,057	59,497	59,560	59,563
	Saving		0.4%	1.5%	16.2%	0.4%	0.3%	0.2%	19.1%
Marine eutrophication	kg N eq	1,257	1,257	1,138	1,257	1,228	1,249	1,237	1,079
	Saving		<0.1%	9.5%	—	2.3%	0.7%	1.6%	15.8%
Marine ecotoxicity—ReCiPe	kg 1,4-DB eq	17,332	17,301	17,047	17,322	17,255	17,290	17,275	16,882
	Saving		0.1%	1.6%	—	0.4%	0.2%	0.3%	2.8%
Marine ecotoxicity—CML 2001	kg 1,4-DB eq	3,607	3,606	3,214	3,608	3,512	3,581	3,541	3,023
	Saving		<0.1%	10.9%	—	2.6%	0.7%	1.8%	18%
Marine ecotoxicity—AUPLCI	day	2.48×10^{-7}	2.47E-07	2.44E-07	2.47E-07	2.47E-07	2.47E-07	2.47E-07	2.43E-07
	Saving		0.1%	1.1%	—	0.3%	0.1%	0.2%	2.0%

5 | SCENARIO RESULTS

The scenarios above collectively constituted an impact reduction strategy to inform planning and implementation of operational decisions by the AAD in the future. The impact assessment results for the different scenarios are reported in Table 3. For climate change potential and CED, the scenario with the largest reduction potential was the introduction of the new icebreaker (15.4% and 16.2% saving, respectively). The building

benchmarking scenario showed potential for the highest savings across the different ecotoxicity indicators (15.1–10.9%, depending on the indicator). The sum of the saving potentials was between 2.0% and 19.1%, depending on the environmental indicator.

6 | DISCUSSION

Validation of the outcomes of this study is challenging, given the limited number of previous studies on Antarctic stations. The baseline modeling identified that the drivers of climate change potential and CED were the use of MGO and SAB for freight movement by the AAD icebreaker, electricity generation and cogenerated heat, on-site vehicles, boilers, and aircraft. This was also a finding by Pertierra et al. (2013), who identified that the major drivers of greenhouse gas emissions of a remote station in Antarctica were transport of people and cargo. Klee reported that electricity generation at Casey contributed to CO₂ emissions of approximately 1,300 tonne per year, which is of similar magnitude to this study (1,486 tonne CO₂ eq, of which 1,462 tonne is from CO₂). However, Klee (2005) excluded transport impacts, and thus underestimated the annual impacts associated with supporting Antarctic research stations by a factor of over three. The greenhouse gas footprint for Casey is equivalent to an impact of 126.3 tonne CO₂ eq per person year.

The dominant process drivers for eutrophication impacts were found not to be those associated with WWT. Rather, they were found to be associated with the use of SAB fuel on station. This finding is significant, as it has highlighted to AAD that the environmental impacts often associated with WWT processes, were not dominated by WWT.

In the case of marine ecotoxicity, this indicator is driven by electricity generation, cogenerated heat, WWT, and the use of on-site vehicles. As reported above, some of the modeled ecotoxicity impacts were premised on heavy metals in wastewater, which were modeled from secondary data sources. This potential for ecotoxicity impacts arising from heavy metals in wastewater was disputed by the AAD, despite there being evidence of the presence of heavy metals wastewater discharged from a similar WWT plant operated by AAD at Davis Station (Stark et al., 2015). The ecotoxicity outcomes indicate the need for improved monitoring and evaluation of wastewater emissions at Casey, and the development of regionally relevant life cycle impact models for Antarctica.

The benefits of different individual scenarios were generally limited to less than a 1% benefit, except for the new icebreaker. Even if these were to be enacted, the lack of data collection would make it difficult to ascertain if the implementation of these scenarios would result in an environmental benefit. The exception to this rule was the new icebreaker scenario, which was identified as providing significant environmental benefits.

7 | LIMITATIONS AND FUTURE RESEARCH OPPORTUNITIES

The lack of consistent, complete, and temporally relevant data was a significant barrier to the completion of a high-quality LCI. We identified several factors which contributed to this, specifically the heavy reliance on human data collection, limited formal processes to capture data suitable for LCA analysis as opposed to data in general, the use of data collection systems inappropriate for Antarctica's environmental conditions, and communication challenges between AAD staff based in Tasmania and those on station with regard to sourcing the data needed for an LCA from station when this is not part of the standard collection and reporting framework.

The environmental impact reductions above are for global and local impacts, depending on the impact category. For climate change potential and CED the impacts are assessed globally, and for marine eutrophication potential and the various ecotoxicity potential measures, the impacts are assessed locally. The need to develop Antarctic specific environmental impact models that incorporates the unique elements of the environmental conditions should be a priority focus of funding jointly by nations present on the continent.

The limitations of this analysis include the difference between the climate zones of the Casey Station buildings and the CBECS benchmark buildings, and the primary building purpose. There exists an opportunity to develop Antarctic specific benchmarks for specific building purposes.

8 | CONCLUSIONS

Over a 6-year timeframe, the AAD embarked on the journey of considering their operations through the lens of LCA. For the first time, the AAD, from the Executive through to Station Leaders and expeditioners on station, can now understand the environmental impacts of their operations across a suite of environmental indicators.

The use of fuel to move cargo and expeditioners and electricity generation on station were key activities driving environmental impacts. Several scenarios incorporating efficiency improvements were also modeled. These scenarios form the basis of an impact reduction strategy for the AAD. The alignment, implications, and opportunities for AAD's and other nation's Antarctic operations will be explored in a subsequent article.

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CONFLICT OF INTEREST

The authors declare no conflict of interest.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.

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